Younes Tatari

Personal Website — younes.tatari@sci.utah.edu — Linkedin — Google Scholar — YouTube

Education

University of Utah **2023-Present (anticipate graduation : March 2027)**
PhD in Mechanical Engineering
Thesis: Computational Modeling of Fluid–Particle Interactions: Multiphysics and Data-driven Approaches

Harbin Institute of Technology **2018-2020**
MS in Energy Science and Engineering
Thesis: Numerical modeling of corona discharge using Finite Volume Method (FVM) in OpenFOAM

Skills

CFD Software OpenFOAM, SimVascular, Ansys-Fluent, Star-CCM
Fluid Physics Biotransport, Multiphase, Lagrangian particle tracking, Heat transfer, Blood flow simulation
Particle Method LIGGGHTS (CFDEM), FlowVC, DPM, MPPIC
Languages Python, Matlab, C++, git
Visualization ParaView, 3D Slicer
Courses FEM, Scientific Machine learning, Fluid Mechanics, Neural Networks
HPC Linux, SLURM
CAD Autodesk Inventor, SolidWorks
ML PyTorch, scikit-learn, TensorFlow, CNNs, PINNs, NumPy, Pandas, SciPy, Streamlit, Agentic AI

Experience

CNH Industrial **May 2026 – Aug 2026**
Soil and Crop Modeling Intern *New Holland, PA, USA*

- Developed a data-driven reduced-order surrogate model (ROM) from CFD simulations, using Proper Orthogonal Decomposition (POD) for dimensionality reduction and Gaussian Process Regression (GPR) for efficient prediction.

Scientific Computing and Imaging Institute at the University of Utah **Aug 2023 – present**
Graduate Research Assistant *Salt Lake City, UT, USA*

- Designed a novel microcatheter for targeted embolization. My design improved delivery efficiency by 10–20% compared to the standard End-Hole Catheter (SEHC) and other commercial designs. ***Provisional Patent***
- Patient Specific Modeling of Hemodynamics During Splenic Artery Embolization.
- CFD simulation of silk-elastinlike protein polymers (SELPs) as liquid embolic egents using Non-Newtonian models in RheoTool within OpenFOAM
- Implemented four-way fluid-particle coupling in successively bifurcating vessels (CFD–DEM).
- Combined Sparse Identification of Nonlinear Dynamical Systems (SINDy) with Stochastic Gradient Descent to overcome SINDy's limitation regarding prior knowledge of candidate terms in the library (ADAM-SINDy).
- Developed data-driven and ML to generate interpretable predictive models for fracture sealing based on CFD-DEM simulation data in geothermal systems.

Harbin Institute of Technology **Aug 2018 – Jun 2020**
Graduate Research Assistant *Harbin, China*

- Investigation of cavitation under the high electric pulse in compressible fluid and electric force calculation.
- Numerical modeling of corona discharge using Finite Volume Method (FVM) in OpenFOAM.
- Facial Keypoints Detection using convolutional neural network and machine learning technics.

Publications

A Dual-Lumen Microcatheter for Minimizing Particle Reflux During Embolization: Proof-of-concept with multiphysics simulations 2026
Computers in Biology and Medicine

Investigation of particle transport in geothermal systems using integrated CFD–DEM and data-driven approaches 2026
Journal of Geothermics

Adam-sindy: An efficient optimization framework for parameterized nonlinear dynamical system identification 2026
Journal of Physical Review Research

Optimizing distal and proximal splenic artery embolization with patient-specific computational fluid dynamics 2024
Journal of Biomechanics